

Hematological anomalies have been observed in patients with prolonged liver damage. For example, patients with chronic cirrhosis (scarring of the liver) often develop anemia and other blood disorders. Hemoglobin is the metalloprotein within red blood cells that is responsible for the oxygen-carrying capacity of blood, and its function may be affected by liver damage. The poor prognosis of patients with cirrhosis and anemia has stimulated further study of hemoglobin function and oxygen saturation.

Blood samples from both cirrhotic patients and healthy individuals at rest and after anaerobic exercise were exposed to varying partial pressures of oxygen. Measurement of oxyhemoglobin dissociation curves (ODCs [Figure 1]) from the blood samples revealed significant functional differences in the hemoglobin from cirrhotic patients relative to control subjects.

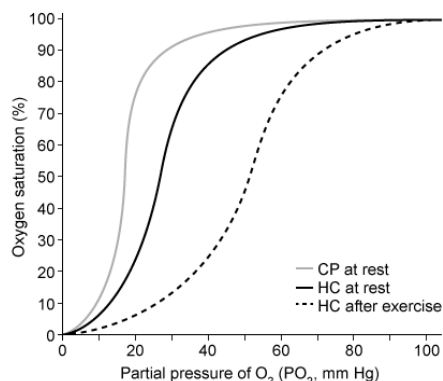


Figure 1 Oxyhemoglobin dissociation curve for cirrhotic patients (CP) at rest and for healthy controls (HC) at rest and after 20 minutes of anaerobic exercise

Vitamins are considered nontoxic supplements and are frequently provided to anemic patients to increase red blood cell production. They may therefore have therapeutic benefits for cirrhotic patients. However, some studies indicate that excessive vitamin consumption may lead to oxidative stress that could induce anemia through hemolysis (red blood cell rupturing). Hemoglobin strongly absorbs light at 577 nm, and researchers conducted hemolysis assays by spinning down red blood cells and measuring absorbance (Abs) of the supernatant at 577 nm to determine how much hemoglobin was released. The assay was carried out in the presence of four vitamins to determine their potential as oxidizing agents, and results are shown in Table 1.

Table 1 Hemolysis of Red Blood Cells from Healthy Donors Treated With Vitamins [2 mM] for 30 Minutes, Measured by Absorbance at 577 nm

Vitamins	Abs 577 nm			
	0 min	10 min	20 min	30 min
B ₁ (thiamine)	0.00	0.02	0.04	0.09
B ₃ (niacin)	0.02	0.04	0.07	0.13
B ₆ (pyridoxine)	0.00	0.06	0.13	0.19
B ₁₂ (cobalamin)	0.01	0.11	0.19	0.24

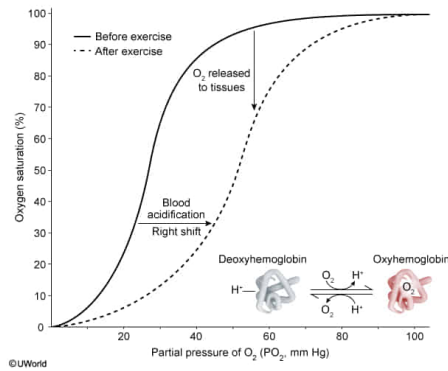
The effect of anaerobic exercise on the oxygen saturation curve in healthy patients is shown in Figure 1. Which of the following is involved in the change in the oxygen dissociation curve?

- ☐ A. CO₂ production decreases in contracting muscle [3%]
- ☐ B. Exercise reduces the PO₂ of oxygen in muscles [20%]
- ☐ C. Hemoglobin binds oxygen tightly during exercise [10%]
- ✓ ☒ D. The blood becomes acidified during exercise [65%]

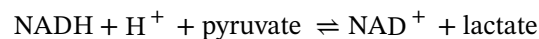
Correct

65% answered correctly

Explanation:



For muscles to contract during exercise, they must **break down glucose** to produce ATP. ATP can then be hydrolyzed to provide the energy for muscle contractions, and this hydrolysis results in the **production of protons**. Glycolysis requires the continuous reduction of NAD⁺ to NADH as glucose is oxidized to pyruvate. Cells must oxidize NADH back to NAD⁺ to continue glycolysis. Under anaerobic conditions, this is accomplished by coupling oxidation of NADH to the reduction of pyruvate, **producing lactate**.



The lactate is then exported into the blood, where it can be transported to the liver to undergo gluconeogenesis. Protons produced during muscle contractions are cotransported with the lactate, causing acidification of the blood.

Therefore, during the 20 minutes of exercise, healthy individuals experience an increase in the H⁺ concentration in the blood. Accumulated H⁺ ions can bind to hemoglobin and reduce its affinity for oxygen in a phenomenon known as the **Bohr effect**. This reduced affinity appears as a **right shift** in the ODC, as shown in Figure 1. The reduction in hemoglobin affinity for oxygen allows more O₂ to be unloaded to the muscle cells. Therefore, the need for more oxygen in contracting muscles is met because the byproducts generated during exercise decrease hemoglobin-oxygen binding.

(Choice A) CO₂ production is increased, not decreased, in contracting muscles. Like lactic acid buildup, this process also contributes to hemoglobin's reduced affinity for oxygen.

(Choice B) Exercise does indeed decrease the partial pressure of oxygen (PO₂) in muscles as oxygen is consumed. However, changes in oxygen partial pressure correspond to different positions *along the ODC*, not to changes in its shape.

(Choice C) Hemoglobin binds oxygen *less* tightly during exercise due to a decrease in blood pH.

Educational objective:

Acidic byproducts of glycolysis produced during exercise cause a right shift in the hemoglobin oxygen dissociation curve. This phenomenon, known as the Bohr effect, provides muscles with a means to receive more oxygen from the blood during activity.

Time spent: 0 sec
Subject:
Biology

QID: 400399
System:
Circulation and Respiration

Hematological anomalies have been observed in patients with prolonged liver damage. For example, patients with chronic cirrhosis (scarring of the liver) often develop anemia and other blood disorders. Hemoglobin is the metalloprotein within red blood cells that is responsible for the oxygen-carrying capacity of blood, and its function may be affected by liver damage. The poor prognosis of patients with cirrhosis and anemia has stimulated further study of hemoglobin function and oxygen saturation.

Blood samples from both cirrhotic patients and healthy individuals at rest and after anaerobic exercise were exposed to varying partial pressures of oxygen. Measurement of oxyhemoglobin dissociation curves (ODCs [Figure 1]) from the blood samples revealed significant functional differences in the hemoglobin from cirrhotic patients relative to control subjects.

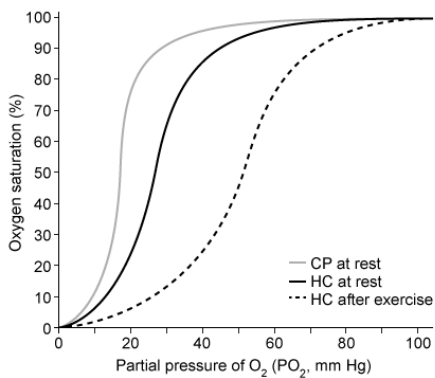


Figure 1 Oxyhemoglobin dissociation curve for cirrhotic patients (CP) at rest and for healthy controls (HC) at rest and after 20 minutes of anaerobic exercise

Vitamins are considered nontoxic supplements and are frequently provided to anemic patients to increase red blood cell production. They may therefore have therapeutic benefits for cirrhotic patients. However, some studies indicate that excessive vitamin consumption may lead to oxidative stress that could induce anemia through hemolysis (red blood cell rupturing). Hemoglobin strongly absorbs light at 577 nm, and researchers conducted hemolysis assays by spinning down red blood cells and measuring absorbance (Abs) of the supernatant at 577 nm to determine how much hemoglobin was released. The assay was carried out in the presence of four vitamins to determine their potential as oxidizing agents, and results are shown in Table 1.

Table 1 Hemolysis of Red Blood Cells from Healthy Donors Treated With Vitamins [2 mM] for 30 Minutes, Measured by Absorbance at 577 nm

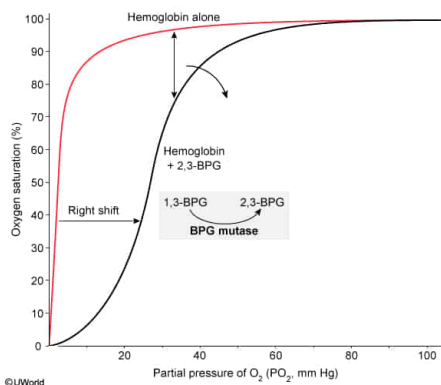
Vitamins	Abs 577 nm			
	0 min	10 min	20 min	30 min
B ₁ (thiamine)	0.00	0.02	0.04	0.09
B ₃ (niacin)	0.02	0.04	0.07	0.13
B ₆ (pyridoxine)	0.00	0.06	0.13	0.19
B ₁₂ (cobalamin)	0.01	0.11	0.19	0.24

Upregulation of which enzyme in red blood cells would best improve oxygen release into the tissues of cirrhotic patients?

- ☐ A. Phosphoglycerate kinase [19%]
- ☐ B. Fructose-2,6-bisphosphatase [33%]
- ☒ C. Bisphosphoglycerate mutase [38%]
- ☐ D. Phosphoglycerate mutase [8%]

Correct
37% answered correctly

Explanation:



Hemoglobin's affinity for oxygen is **decreased** in the presence of **2,3-bisphosphoglycerate** (2,3-BPG) within the red blood cell. 2,3-BPG binds hemoglobin at an **allosteric site** and stabilizes the deoxygenated conformation (T state). This reduces oxygen binding and increases O_2 release into tissues (right shift of oxygen dissociation curve). Therefore, upregulation of an enzyme that produces 2,3-BPG could **improve oxygen release**.

Mutases facilitate reactions that change the location of a functional group within a molecule. The enzyme **bisphosphoglycerate mutase** converts 1,3-BPG to 2,3-BPG by transferring a phosphate from carbon 1 to carbon 2 within a bisphosphoglycerate molecule. Upregulation of this enzyme would therefore increase the concentration of 2,3-BPG and could facilitate oxygen release into the tissues of cirrhotic patients.

(Choice A) Kinases add and, in some cases, remove phosphate groups. **Phosphoglycerate kinase** interconverts 1,3-bisphosphoglycerate and 3-phosphoglycerate and is not involved in production of 2,3-BPG.

(Choice B) Fructose-2,6-bisphosphatase removes a phosphate group from fructose-2,6-bisphosphate to produce fructose-6-phosphate. This reaction helps regulate glycolysis and gluconeogenesis but does not directly affect the affinity of hemoglobin for oxygen.

(Choice D) Although **phosphoglycerate mutase** does have mutase activity, it interconverts 3-phosphoglycerate and 2-phosphoglycerate. It acts on *phosphoglycerates*, not *bisphosphoglycerates*.

Educational objective:

Bisphosphoglycerate mutase converts 1,3-bisphosphoglycerate into 2,3-bisphosphoglycerate, which allosterically regulates hemoglobin. It decreases oxygen affinity by stabilizing the deoxyhemoglobin conformation. Increasing concentrations of 2,3-bisphosphoglycerate produce a right shift in the oxyhemoglobin dissociation curve and favor oxygen delivery to tissues.

Time spent: 0 sec
Subject:
Biology

QID: 400400
System:
Circulation and Respiration

Hematological anomalies have been observed in patients with prolonged liver damage. For example, patients with chronic cirrhosis (scarring of the liver) often develop anemia and other blood disorders. Hemoglobin is the metalloprotein within red blood cells that is responsible for the oxygen-carrying capacity of blood, and its function may be affected by liver damage. The poor prognosis of patients with cirrhosis and anemia has stimulated further study of hemoglobin function and oxygen saturation.

Blood samples from both cirrhotic patients and healthy individuals at rest and after anaerobic exercise were exposed to varying partial pressures of oxygen. Measurement of oxyhemoglobin dissociation curves (ODCs [Figure 1]) from the blood samples revealed significant functional differences in the hemoglobin from cirrhotic patients relative to control subjects.

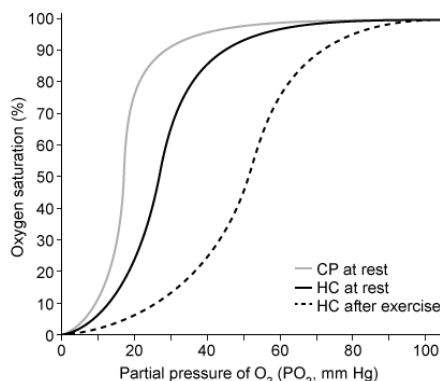


Figure 1 Oxyhemoglobin dissociation curve for cirrhotic patients (CP) at rest and for healthy controls (HC) at rest and after 20 minutes of anaerobic exercise

Vitamins are considered nontoxic supplements and are frequently provided to anemic patients to increase red blood cell production. They may therefore have therapeutic benefits for cirrhotic patients. However, some studies indicate that excessive vitamin consumption may lead to oxidative stress that could induce anemia through hemolysis (red blood cell rupturing). Hemoglobin strongly absorbs light at 577 nm, and researchers conducted hemolysis assays by spinning down red blood cells and measuring absorbance (Abs) of the supernatant at 577 nm to determine how much hemoglobin was released. The assay was carried out in the presence of four vitamins to determine their potential as oxidizing agents, and results are shown in Table 1.

Table 1 Hemolysis of Red Blood Cells from Healthy Donors Treated With Vitamins [2 mM] for 30 Minutes, Measured by Absorbance at 577 nm

Vitamins	Abs 577 nm			
	0 min	10 min	20 min	30 min
B ₁ (thiamine)	0.00	0.02	0.04	0.09
B ₃ (niacin)	0.02	0.04	0.07	0.13
B ₆ (pyridoxine)	0.00	0.06	0.13	0.19
B ₁₂ (cobalamin)	0.01	0.11	0.19	0.24

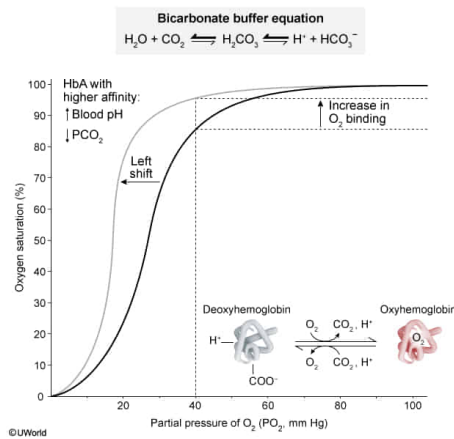
Hyperventilation in healthy patients leads to oxygen dissociation curves similar to those seen in patients with cirrhosis when at rest. Based on this information, what physiological change in cirrhotic patients most likely decreases the level of oxygen delivery to tissues?

- ✔ ☒ A. Low carbonic acid and CO₂ in the blood [55%]
- ☐ B. Increased excretion of bicarbonate by kidneys [14%]
- ☐ C. Higher PO₂ in the lungs of cirrhotic patients [9%]
- ☐ D. Lower affinity of hemoglobin for oxygen in tissues [20%]

Correct

55% answered correctly

Explanation:



The excessive exhalation that occurs during hyperventilation **depletes blood CO₂**. Carbon dioxide is in equilibrium with carbonic acid (H₂CO₃), which in turn is in equilibrium with bicarbonate (HCO₃⁻) in the blood.



This equilibrium forms the basis for the bicarbonate buffering system, which helps control blood pH. As CO₂ is removed, the equilibrium shifts to favor production of more CO₂ via the **consumption of carbonic acid and bicarbonate**, which increases blood pH (fewer H⁺ ions). Both CO₂ and H⁺ allosterically decrease the affinity of hemoglobin for oxygen, so their removal will *increase* the affinity and result in **less delivery of oxygen to tissues** as hemoglobin fails to release it. An increase in oxygen affinity appears as a **left shift** in the ODC.

Like hyperventilating individuals, patients with cirrhosis exhibit a left shift in their ODC relative to healthy subjects. It is therefore reasonable to hypothesize that patients with cirrhosis experience decreased carbonic acid and/or CO₂ levels.

(Choice B) Bicarbonate is the conjugate base of carbonic acid. When the kidneys excrete bicarbonate, they remove bicarbonate from the blood. This shifts both equilibria toward the right, producing carbonic acid from CO₂ in the blood followed by dissociation of the carbonic acid to form bicarbonate and a proton. The production of protons lowers the blood pH, resulting in a right shift of the ODC.

(Choice C) There is no indication in the passage that patients with cirrhosis have lung abnormalities. Therefore, the partial pressure of oxygen (PO₂) in the lungs of these patients is most likely the same as in healthy controls. In addition, the partial pressure of oxygen is used to *create* the ODC; it does not shift it.

(Choice D) A decrease in hemoglobin oxygen affinity would correspond to a *right shift* of the ODC and more oxygen being released into tissue.

Educational objective:

A left shift in the oxyhemoglobin dissociation curve signifies an increase in the affinity of hemoglobin for oxygen. This affinity is inversely related to CO₂ and H⁺ concentrations. Therefore, a left shift in the ODC often indicates lower CO₂ and/or acid levels in the blood.

Time spent: 0 sec
Subject:
Biology

QID: 400401
System:
Circulation and Respiration

Hematological anomalies have been observed in patients with prolonged liver damage. For example, patients with chronic cirrhosis (scarring of the liver) often develop anemia and other blood disorders. Hemoglobin is the metalloprotein within red blood cells that is responsible for the oxygen-carrying capacity of blood, and its function may be affected by liver damage. The poor prognosis of patients with cirrhosis and anemia has stimulated further study of hemoglobin function and oxygen saturation.

Blood samples from both cirrhotic patients and healthy individuals at rest and after anaerobic exercise were exposed to varying partial pressures of oxygen. Measurement of oxyhemoglobin dissociation curves (ODCs [Figure 1]) from the blood samples revealed significant functional differences in the hemoglobin from cirrhotic patients relative to control subjects.

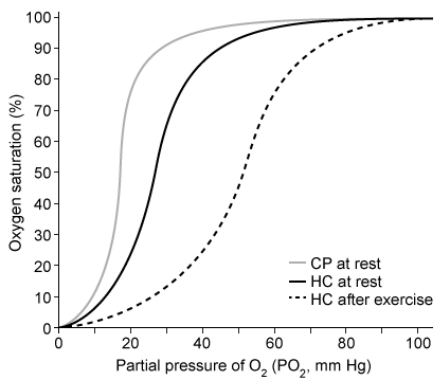


Figure 1 Oxyhemoglobin dissociation curve for cirrhotic patients (CP) at rest and for healthy controls (HC) at rest and after 20 minutes of anaerobic exercise

Vitamins are considered nontoxic supplements and are frequently provided to anemic patients to increase red blood cell production. They may therefore have therapeutic benefits for cirrhotic patients. However, some studies indicate that excessive vitamin consumption may lead to oxidative stress that could induce anemia through hemolysis (red blood cell rupturing). Hemoglobin strongly absorbs light at 577 nm, and researchers conducted hemolysis assays by spinning down red blood cells and measuring absorbance (Abs) of the supernatant at 577 nm to determine how much hemoglobin was released. The assay was carried out in the presence of four vitamins to determine their potential as oxidizing agents, and results are shown in Table 1.

Table 1 Hemolysis of Red Blood Cells from Healthy Donors Treated With Vitamins [2 mM] for 30 Minutes, Measured by Absorbance at 577 nm

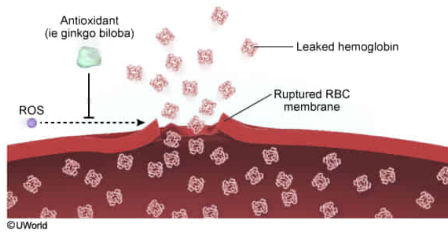
Vitamins	Abs 577 nm			
	0 min	10 min	20 min	30 min
B ₁ (thiamine)	0.00	0.02	0.04	0.09
B ₃ (niacin)	0.02	0.04	0.07	0.13
B ₆ (pyridoxine)	0.00	0.06	0.13	0.19
B ₁₂ (cobalamin)	0.01	0.11	0.19	0.24

A patient treated for cirrhosis was given vitamin B₆ to stimulate red blood cell production. To counteract the potential negative side effects of this treatment, extract of ginkgo biloba was also administered. What is the most likely role of ginkgo biloba extract in this scenario?

- ☐ A. It stimulates hemoglobin synthesis [6%]
- ☐ B. It increases hemoglobin's oxygen affinity [5%]
- ☒ C. It neutralizes reactive oxygen species [80%]
- ☐ D. It inhibits blood acidification [7%]

Correct
80% answered correctly

Explanation:



The passage states that although vitamins may help induce red blood cell production, they can also cause oxidative stress that results in **hemolysis**.

Oxidative stress occurs when **reactive oxygen species** (ROS) such as hydrogen peroxide are overly abundant near cell membranes. ROS can react with lipids in cell membranes and cause membrane ruptures, also known as lysis. When this occurs in red blood cells, hemoglobin can leak out and denature, becoming nonfunctional and causing hemolytic anemia. This negative side effect would therefore be best treated by a molecule that **neutralizes ROS**, also known as an **antioxidant**. Ginkgo biloba extract is therefore most likely acting as an antioxidant to remove ROS from the system.

(Choice A) Hemoglobin synthesis is already stimulated by vitamin B₆ as more red blood cells are produced.

Underproduction of hemoglobin is not the side effect that needs to be counteracted.

(Choice B) An increase in hemoglobin's oxygen affinity would not help reduce oxidative stress.

(Choice D) Blood pH involves acid-base chemistry. Oxidative stress is related to oxidation-reduction reactions.

Educational objective:

Oxidative stress, caused by reactive oxygen species, can cause hemolysis. Compounds that remove reactive oxygen species can relieve the adverse effects of oxidative stress.

Time spent: 0 sec

Subject:

Biology

QID: 400402

System:

Circulation and Respiration

Hematological anomalies have been observed in patients with prolonged liver damage. For example, patients with chronic cirrhosis (scarring of the liver) often develop anemia and other blood disorders. Hemoglobin is the metalloprotein within red blood cells that is responsible for the oxygen-carrying capacity of blood, and its function may be affected by liver damage. The poor prognosis of patients with cirrhosis and anemia has stimulated further study of hemoglobin function and oxygen saturation.

Blood samples from both cirrhotic patients and healthy individuals at rest and after anaerobic exercise were exposed to varying partial pressures of oxygen. Measurement of oxyhemoglobin dissociation curves (ODCs [Figure 1]) from the blood samples revealed significant functional differences in the hemoglobin from cirrhotic patients relative to control subjects.

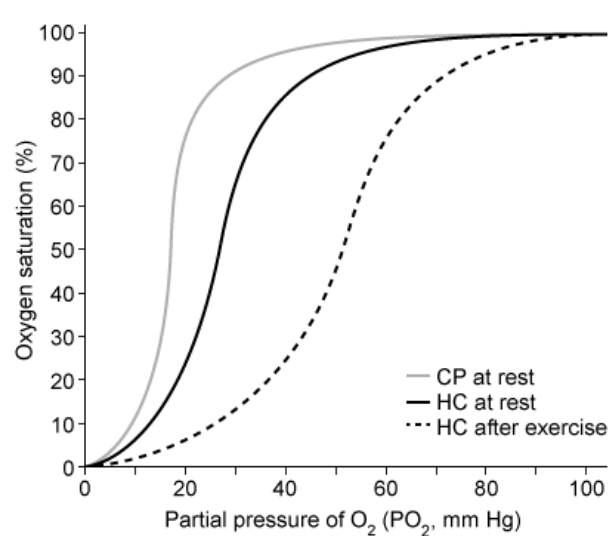


Figure 1 Oxyhemoglobin dissociation curve for cirrhotic patients (CP) at rest and for healthy controls (HC) at rest and after 20 minutes of anaerobic exercise

Vitamins are considered nontoxic supplements and are frequently provided to anemic patients to increase red blood cell production. They may therefore have therapeutic benefits for cirrhotic patients. However, some studies indicate that excessive vitamin consumption may lead to oxidative stress that could induce anemia through hemolysis (red blood cell rupturing). Hemoglobin strongly absorbs light at 577 nm, and researchers conducted hemolysis assays by spinning down red blood cells and measuring absorbance (Abs) of the supernatant at 577 nm to determine how much hemoglobin was released. The assay was carried out in the presence of four vitamins to determine their potential as oxidizing agents, and results are shown in Table 1.

Table 1 Hemolysis of Red Blood Cells from Healthy Donors Treated With Vitamins [2 mM] for 30 Minutes, Measured by Absorbance at 577 nm

Vitamins	Abs 577 nm			
	0 min	10 min	20 min	30 min
B ₁ (thiamine)	0.00	0.02	0.04	0.09
B ₃ (niacin)	0.02	0.04	0.07	0.13
B ₆ (pyridoxine)	0.00	0.06	0.13	0.19
B ₁₂ (cobalamin)	0.01	0.11	0.19	0.24

Healthy subjects show a shallow increase in oxygen-carrying capacity at 0-20 mm Hg, but a sharp increase at 20-50 mm Hg. Which statements explain this effect?

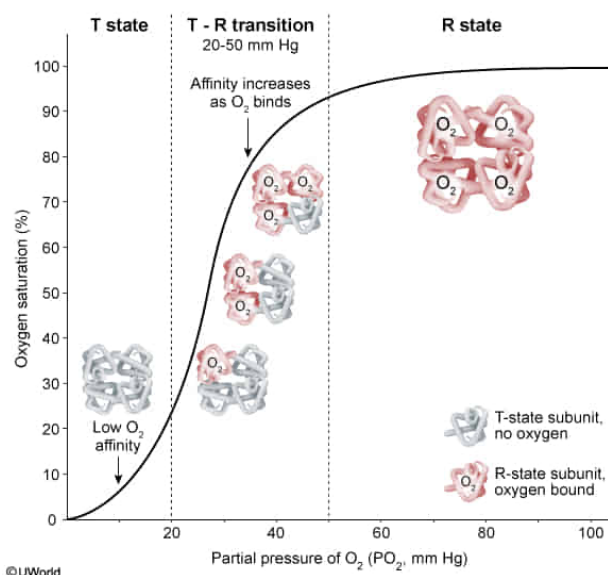
- I. Oxygen binding to a subunit induces a change from the R state to the T state
- II. Hemoglobin affinity for O₂ increases as O₂ pressure increases from 20 to 50 mm Hg
- III. Hemoglobin exhibits positive binding cooperativity
- IV. Hemoglobin exhibits negative binding cooperativity

- ☐ A. I and IV [1%]
- ☒ B. II and III [55%]
- ☐ C. I, II, and III [40%]
- ☐ D. I, II, and IV [1%]

Correct

54% answered correctly

Explanation:



Cooperativity is the phenomenon in which binding of a ligand to a biomolecule

alters the affinity of that biomolecule for additional ligands. **Positive cooperativity** occurs when binding of one ligand **increases the affinity** for additional ligands, and **negative cooperativity** occurs when binding of one ligand **decreases the affinity** for additional ligands. Generally, a protein must have multiple subunits or binding sites to exhibit cooperativity.

Hemoglobin consists of four subunits, each of which can exist in either of two conformations: the T state ("tense," low oxygen affinity) or the R state ("relaxed," high oxygen affinity). Oxygen's binding to one subunit stabilizes the subunit's R state. The bound subunit then "positively cooperates" with other subunits, **inducing them to change** from the T to the R conformation and making additional oxygen binding easier. This is shown on the graph of binding versus O₂ partial pressure by the **sigmoidal (S-shaped) curve**. The curve steepens as O₂ partial pressure increases from 20 mm Hg to 50 mm Hg. Therefore, overall **oxygen affinity increases with O₂ pressure** within this range (**Number II**), and hemoglobin exhibits positive binding cooperativity (**Number III**).

(Number I) Oxygen binding induces a change from the T state to the R state, not from R to T.

(Number IV) Negative binding cooperativity would result if oxygen binding to one subunit stabilized the T state of other subunits. Many cell receptors, for example, exhibit negative cooperativity for their ligands. In **graphical form**, this would be reflected as a relatively steep initial binding curve followed by an abrupt change to a shallower slope.

Educational objective:

Hemoglobin's affinity for oxygen increases as successive oxygen molecules bind to individual subunits and change neighboring subunits from a tense (T) to a relaxed (R) state. Positive cooperativity between binding sites in proteins and enzymes is indicated by the sigmoidal (S) shape of kinetic graphs.

Time spent:0 sec

Subject:
Biology

QID:400403

System:
Circulation and Respiration

Hematological anomalies have been observed in patients with prolonged liver damage. For example, patients with chronic cirrhosis (scarring of the liver) often develop anemia and other blood disorders. Hemoglobin is the metalloprotein within red blood cells that is responsible for the oxygen-carrying capacity of blood, and its function may be affected by liver damage. The poor prognosis of patients with cirrhosis and anemia has stimulated further study of hemoglobin function and oxygen saturation.

Blood samples from both cirrhotic patients and healthy individuals at rest and after anaerobic exercise were exposed to varying partial pressures of oxygen. Measurement of oxyhemoglobin dissociation curves (ODCs [Figure 1]) from the blood samples revealed significant functional differences in the hemoglobin from cirrhotic patients relative to control subjects.

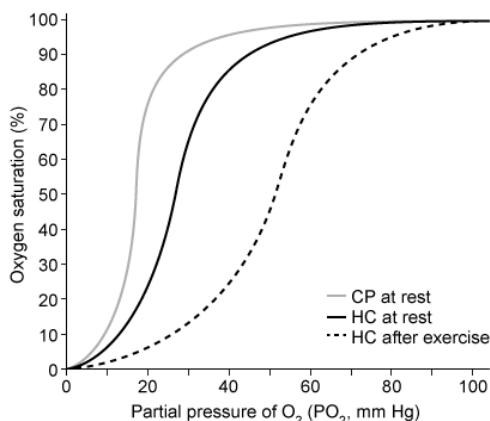


Figure 1 Oxyhemoglobin dissociation curve for cirrhotic patients (CP) at rest and for healthy controls (HC) at rest and after 20 minutes of anaerobic exercise

Vitamins are considered nontoxic supplements and are frequently provided to anemic patients to increase red blood cell production. They may therefore have therapeutic benefits for cirrhotic patients. However, some studies indicate that excessive vitamin consumption may lead to oxidative stress that could induce anemia through hemolysis (red blood cell rupturing). Hemoglobin strongly absorbs light at 577 nm, and researchers conducted hemolysis assays by spinning down red blood cells and measuring absorbance (Abs) of the supernatant at 577 nm to determine how much hemoglobin was released. The assay was carried out in the presence of four vitamins to determine their potential as oxidizing agents, and results are shown in Table 1.

Table 1 Hemolysis of Red Blood Cells from Healthy Donors Treated With Vitamins [2 mM] for 30 Minutes, Measured by Absorbance at 577 nm

Vitamins	Abs 577 nm			
	0 min	10 min	20 min	30 min
B ₁ (thiamine)	0.00	0.02	0.04	0.09
B ₃ (niacin)	0.02	0.04	0.07	0.13
B ₆ (pyridoxine)	0.00	0.06	0.13	0.19
B ₁₂ (cobalamin)	0.01	0.11	0.19	0.24

Individuals with glucose-6-phosphate dehydrogenase (G6PD) deficiencies are more susceptible to oxidative stress than others. If a patient with cirrhosis also suffered from G6PD deficiency, which vitamin could cause the most severe reduction in that patient's hematocrit?

- ☐ A. Vitamin B₁ [20%]
- ☐ B. Vitamin B₃ [3%]

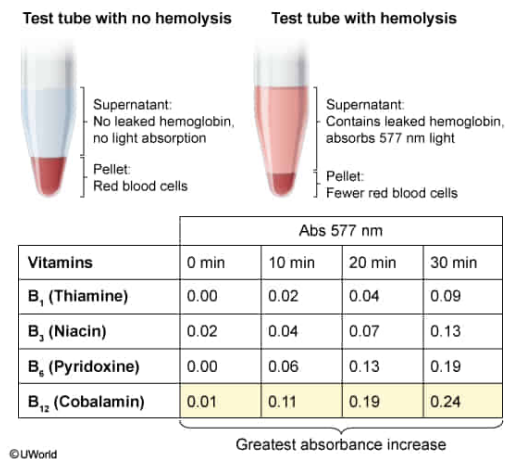
☐ C. Vitamin B₆ [5%]

✓ ☒ D. Vitamin B₁₂ [70%]

Correct

69% answered correctly

Explanation:



The passage states that vitamins are generally nontoxic but may cause some oxidative stress. In patients with increased susceptibility to oxidative stress, administration of vitamin supplements could result in hemolysis. Of the vitamins measured in Table 1, **vitamin B₁₂ (cobalamin)** displays the **greatest rate of hemolysis**.

Hematocrit is a standard laboratory measurement of red blood cell volume as a **percentage of total blood volume**. It indicates the **amount of intact red blood cells** within a blood sample. An increase in absorbance at 577 nm in the supernatant indicates hemoglobin leakage and, therefore, hemolysis. Hemolysis would cause a decrease in hematocrit. The absorbance at 577 nm increases most quickly with administration of vitamin B₁₂, which would therefore produce the greatest net reduction of hematocrit.

(Choices A, B, and C) Vitamins B₁, B₃, and B₆ may decrease hematocrit through hemolysis, but not as quickly as vitamin B₁₂.

Educational objective:

Hematocrit, measured as a percentage of the total blood volume, indicates the amount of red blood cells in a blood sample. Hemolysis induced by oxidative stress can reduce hematocrit.

Time spent: 0 sec
Subject:
Biology

QID: 400404
System:
Circulation and Respiration